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Software Requirements Specification (SRS)

Abyss Navigator Suite Project

Brigman Deep Sea Technology

March 5, 2026

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Revision History

Version	Date	Author	Reason for Changes
0.1	March 5, 2026	BDST Requirements Team	Initial SRS draft structured using the Karl E. Wiegers SRS template (1999) and IEEE 830-aligned categorization.

1 Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the software requirements for the **Abyss Navigator Suite**, an unmanned deep-sea submersible fleet control system for Triton Deepwater. It is intended to be the baseline agreement for design, implementation, verification, and change control.

1.2 Document Conventions

Template source. This SRS is structured using the *Software Requirements Specification* template by **Karl E. Wiegers (1999)** (provided in *References/srs_template-ieee.pdf*), adapted to incorporate the course rubric requirements (requirements tree, codification, iRTM, and change mitigation).

Normative language. All requirements use **shall** to indicate mandatory behavior.

Codification (mandatory). The following ID schemes are used throughout:

- **Software requirements:** <DOMAIN>-<FEATURE>-<NNN> (e.g., SAF-EMG-001).
- **Technology requirements:** TECH-<AREA>-<NNN> (e.g., TECH-PLAT-001).
- **Quality model features:** QUAL-<AREA>-<NNN> (e.g., QUAL-SAF-001).
- **iRTM entries:** RTM-<NNN> (e.g., RTM-001).
- **Change mitigation controls:** CHG-<AREA>-<NNN> (e.g., CHG-CTRL-001).

Priority levels. Priority is recorded as **Essential**, **Desirable**, or **Acceptable** to align with the Project Specification.

1.3 Intended Audience and Reading Suggestions

This SRS is intended for Triton Deepwater stakeholders, the Brigman development and verification teams, project management, and independent auditors. Readers should begin with Sections 1–3 (scope, constraints, requirements tree and codification) before using Section 4 (requirements) and Sections 7–8 (traceability and change mitigation).

1.4 Product Scope

The Abyss Navigator Suite enables unmanned missions executed autonomously, remotely supervised from a surface or mothership environment, with controlled intervention, safe emergency handling, and operational traceability.

1.5 References

- Triton Deepwater, *Project Specification: The Abyss Navigator Suite*, ICT2113, 2026.
- Brigman Deep Sea Technology, *Software Requirements Analysis Report (BCIC Methodology)*, 2026.
- Kevin Anthony Jones, *Labtorial Guide for Class Project: Reqs now & near future*, Dec 2025.
- Kevin Anthony Jones, *Guide Rubric for Assessment of Deliverables*, Jan 2026.
- Karl E. Wiegers, *Software Requirements Specification Template*, 1999.

2 Overall Description

2.1 Product Perspective

The Abyss Navigator Suite is a new software product that coordinates multiple unmanned submersibles and interfaces with Triton-provided mission sensors, communications links, and external reporting/analysis

environments.

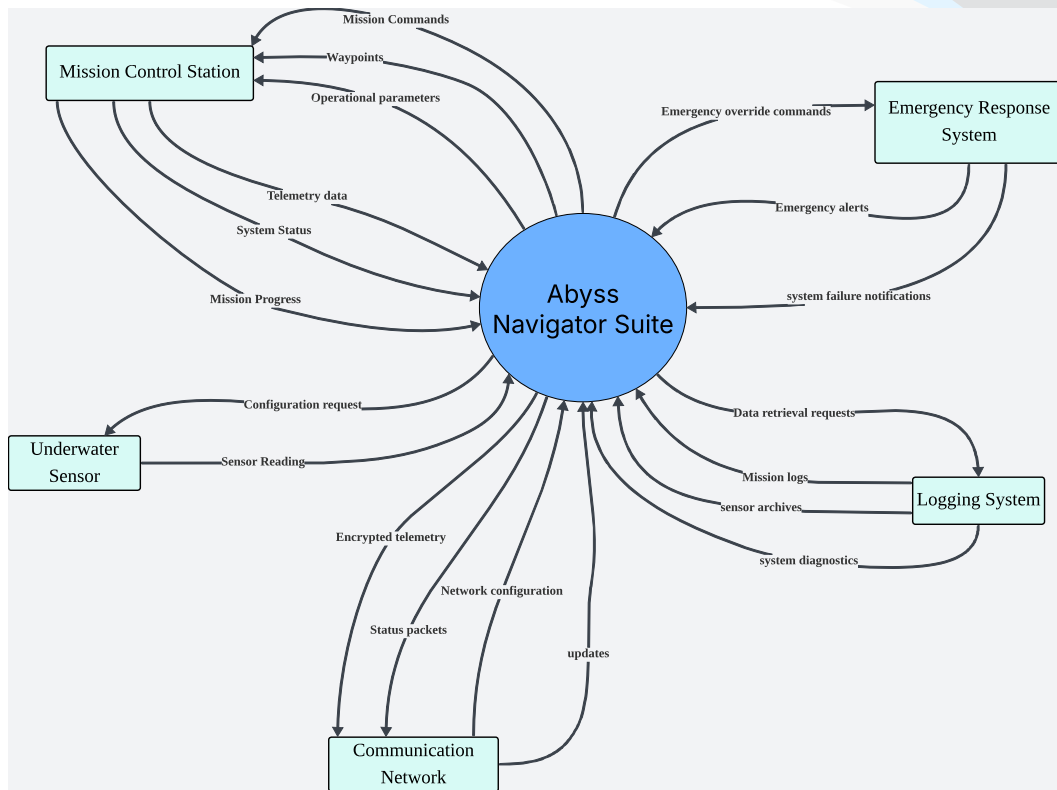


Figure 1: Context Model: Abyss Navigator Suite operating environment and external actors

2.2 Product Functions

At a high level, the system provides:

- Mission planning and autonomous execution for unmanned deep-sea missions.
- Remote monitoring, operator alerting, and controlled remote intervention.
- Fleet-level supervision for multiple submersibles concurrently.
- Emergency detection, safe-state transition, and recovery coordination.
- Secure operational data management, logging, and post-mission traceability.

2.3 User Classes and Characteristics

- **Mission Supervisor:** monitors mission progress and authorizes intervention.
- **Fleet Operator:** manages multi-vehicle assignments and scheduling.
- **Safety & Compliance Officer:** reviews anomalies, audits logs, and ensures governance obligations are met.
- **System Administrator:** manages credentials, configuration, and system health.
- **Data Analyst:** exports mission data for downstream analysis.

2.4 Operating Environment (Latest Tech)

The operating environment is characterized by remote supervision from a mothership environment and constrained connectivity to submersibles. The vendor-selected software platform is described in Section 5. Key operating constraints include:

- Underwater communications may be intermittent and low-bandwidth.

- Extreme pressure and low visibility increase reliance on sensor-driven autonomy.
- Operational data is safety- and audit-relevant and must be protected.

2.5 Design and Implementation Constraints

The following constraints apply:

- **Remote operations context:** mission supervision and control are performed remotely from surface/mothership environments.
- **Communications constraints:** the system must operate under constrained bandwidth/latency and intermittent connectivity.
- **Security and governance:** operational data protection and auditability are required.
- **Safety-first policy:** failures and abnormal conditions must bias toward safe-state behavior.

2.6 User Documentation

The following documentation shall be delivered with the software:

- Operator manual (mission planning, supervision, interventions, safety procedures).
- System administration guide (credentials, configuration, deployment).
- Interface/API guide for integrating Triton-provided sensors and external systems.

2.7 Assumptions and Dependencies

- Triton provides mission sensors and their interface specifications for integration.
- Missions are supervised by trained and authorized personnel.
- Regulatory obligations apply to maritime safety and data governance.

3 Requirements Tree & Codification

3.1 Requirements Tree (SRS Organization)

This SRS uses a CRaM-aligned requirements tree that differs from the Project Specification tree by organizing requirements for implementation and verification under software-centric domains:

- **Navigation & Mission Autonomy (NAV)**
- **Communications & Remote Operations (COM)**
- **Safety & Recovery (SAF/SYS)**
- **Data Logging & Accountability (DATA)**
- **External Interfaces (INT)**
- **Security & Governance (SEC)**

Tree Node	Contained Requirement IDs
NAV	NAV-PLAN-001, NAV-EXEC-001, NAV-WPT-001, NAV-FLEET-001
COM	COM-MON-001, COM-INT-001, COM-DEG-001
SAF/SYS	SAF-EMG-001, SAF-EMG-002, SAF-EMG-003, SAF-STATE-001, SYS-CONT-001, SYS-TERM-001, SYS-REC-001
DATA	DATA-LOG-001, DATA-LOG-002, DATA-LOG-003, DATA-LOG-004
INT	INT-SENS-001
SEC	SEC-ENC-001, SEC-AUTH-001, SEC-AUD-001

Table 1: Requirements tree nodes and congruous allocation of SRS requirements

3.2 Codification Rules

Requirement IDs are stable across revisions. The numeric suffix increments only when a new requirement is added; edited requirements retain their IDs to preserve traceability in the iRTM.

3.3 Requirement Attributes and Quality Tagging

Each requirement in Section 4 is recorded with:

- **Source:** originating Project Specification clause(s) (e.g., B0-L1-4, IT-L0-C3).
- **Priority:** Essential / Desirable / Acceptable.
- **Quality tags:** one or more QUAL-* IDs from Section 6.
- **Verification:** inspection, analysis, test, or demonstration.
- **Use Case link:** reference to the primary use case(s) in Figure 2 (numbered UC01, UC02, ...) that realize or exercise the requirement.

4 Specific Requirements

4.1 Navigation & Mission Autonomy

Req ID	Priority	Source	Quality Tags	Requirement	Verification	Use Case Link
NAV-PLAN-001	Essential	BO-L1-1	QUAL-OPS-001, QUAL-TRC-001	The system shall allow an authorized operator to create and version a mission plan containing an ordered set of waypoints (latitude, longitude, depth) and mission phases.	Demo: Create/edit/version plan; inspect stored plan versions.	UC01 (Mission planning)
NAV-EXEC-001	Essential	BO-L1-1, BO-L0-C1	QUAL-RES-001, QUAL-SAF-001	The system shall execute a validated mission plan autonomously without continuous human control for a full mission cycle in a representative simulation.	Test: End-to-end mission simulation with no operator commands.	UC02 (Autonomous mission execution)
NAV-WPT-001	Essential	BO-L1-1	QUAL-OPS-001	The system shall achieve 99% waypoint arrival success in which each reached waypoint is within ≤ 5 m horizontal and ≤ 2 m vertical error under representative sea-trial conditions.	Analysis: Compare logged track vs. waypoints across trials.	UC02 (Waypoint navigation)
NAV-FLEET-001	Desirable	BO-L1-3, IT-L1-3	QUAL-OPS-001, QUAL-RES-001	The system shall support concurrent supervision of at least 3 submersibles, including assignment of mission plans and real-time status visibility per vehicle.	Test: Simulated 3-vehicle session; verify assignment and visibility.	UC03 (Supervise multiple vehicles)

4.2 Communications & Remote Operations

Req ID	Priority	Source	Quality Tags	Requirement	Verification	Use Case Link
COM-MON-001	Essential	BO-L1-2, IT-L1-1	QUAL-OPS-001	The system shall provide remote mission and vehicle status telemetry updates at least once every 30 seconds when connectivity is available.	Test: Measure telemetry update intervals under nominal link.	UC04 (Monitor telemetry)
COM-INT-001	Essential	IT-L1-2, BO-L1-2	QUAL-OPS-001, QUAL-SAF-001	The system shall allow an authorized operator to issue remote intervention commands (pause, resume, safe-state) and receive acknowledgement within ≤ 60 seconds when the link latency is ≤ 30 seconds.	Test: Command/ack timing under simulated 30s latency.	UC05 (Remote intervention)
COM-DEG-001	Essential	IT-L0-C2, IT-L1-6	QUAL-RES-001, QUAL-SAF-001	The system shall support safe operation under constrained communications, including operation at ≥ 1 kbps bandwidth, ≤ 30 s one-way latency, and intermittent connectivity, and shall enter safe-state if connectivity is lost continuously for > 10 minutes.	Test: Fault-inject bandwidth, latency, and outage; verify safe-state trigger.	UC06 (Handle degraded comms)

4.3 Safety & Recovery

Req ID	Priority	Source	Quality Tags	Requirement	Verification	Use Case Link
SAF-EMG-001	Essential	BO-L1-4, BO-L2-4.1	QUAL-SAF-001, QUAL-OPS-001	The system shall detect pressure deviations $> \pm 5\%$ from nominal setpoint within 5 seconds .	Test: Inject simulated pressure spike; measure detection latency.	UC07 (Detect anomalies)
SAF-EMG-002	Essential	BO-L1-4, BO-L2-4.1	QUAL-SAF-001, QUAL-OPS-001	The system shall detect temperature deviations $> \pm 10\%$ from nominal within 10 seconds .	Test: Thermal chamber simulation or sensor-in-the-loop test.	UC07 (Detect anomalies)
SAF-EMG-003	Essential	BO-L1-4, BO-L2-4.1	QUAL-SAF-001, QUAL-OPS-001	The system shall detect power supply deviations $> \pm 15\%$ from nominal within 10 seconds .	Test: Power supply fault injection; measure detection latency.	UC07 (Detect anomalies)
SAF-STATE-001	Essential	BO-L2-4.2	QUAL-SAF-001	Upon detection of any abnormal condition, the system shall transition the submersible to a safe operational state defined as neutral buoyancy, minimal power consumption, and active communication beacon.	Integration test: End-to-end emergency scenario.	UC08 (Enter safe state)

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Req ID	Priority	Source	Quality Tags	Requirement	Verification	Use Case Link
SYS-CONT-001	Essential	BO-L1-5	QUAL-RES-001, QUAL-SAF-001	When non-critical degradation occurs, the system shall continue the mission in a degraded mode while alerting the operator and logging the degradation event.	Test: Inject non-critical fault; confirm degraded continuation and alert/log.	UC09 (Continue in degraded mode)
SYS-TERM-001	Essential	BO-L1-5	QUAL-SAF-001, QUAL-TRC-001	When termination is required, the system shall terminate the mission in an orderly manner by completing the current operation when safe, logging the shutdown reason, and entering safe-state.	Test: Inject critical fault; verify ordered termination sequence.	UC10 (Terminate mission)
SYS-REC-001	Essential	BO-L1-5	QUAL-SAF-001, QUAL-OPS-001	After entering safe-state, the system shall support recovery coordination by continuously broadcasting a communication beacon and exposing last-known position and status to operators until recovery is confirmed.	Demo/Test: Simulate safe-state; verify beacon and status visibility.	UC11 (Coordinate recovery)

4.4 Data Logging & Accountability

Req ID	Priority	Source	Quality Tags	Requirement	Verification	Use Case Link
DATA-LOG-001	Essential	BO-L1-6, IT-L1-5	QUAL-TRC-001	The system shall log mission events and operator actions with timestamp (UTC), vehicle ID, and operator ID attribution.	Inspection/Test: Review log records for required fields.	UC12 (Record operations)
DATA-LOG-002	Essential	IT-L1-5, IT-L0-C3	QUAL-SEC-001, QUAL-TRC-001	The system shall store operational logs in an append-only form and retain them for at least 5 years to support audit and investigation.	Inspection: Verify storage configuration and retention policy.	UC12 (Record operations)
DATA-LOG-003	Essential	BO-L2-6.2, IT-L2-5.1	QUAL-OPS-001, QUAL-TRC-001	The system shall allow authorized retrieval and export of mission records for a ≤ 24 -hour mission into a structured format (CSV and JSON) within ≤ 2 minutes.	Test: Export timed retrieval under representative data volume.	UC13 (Retrieve/export records)
DATA-LOG-004	Desirable	BO-L1-7, BO-L2-6.2	QUAL-OPS-001	The system shall generate a post-mission summary report (mission timeline, anomalies, interventions) within ≤ 10 minutes after mission completion.	Test: Complete mission; measure report generation time and contents.	UC14 (Post-mission summary)

4.5 External Interface Requirements

Req ID	Priority	Source	Quality Tags	Requirement	Verification	Use Case Link
INT-SENS-001	Essential	IT-L0-A1	QUAL-RES-001	The system shall integrate customer-supplied mission sensors using ROS2 middleware and DDS transport, and shall ingest sensor messages in a documented JSON schema defined in the Interface/API guide.	Test: Sensor-in-the-loop integration against the published schema.	UC15 (Sensor integration)

4.6 Security & Governance

Req ID	Priority	Source	Quality Tags	Requirement	Verification	Use Case Link
SEC-ENC-001	Essential	IT-L0-C3, IT-L1-4	QUAL-SEC-001	The system shall protect operational data using AES-256 encryption for data at rest and for encrypted payloads in transit.	Inspection/Test: Verify crypto configuration and encrypted traffic capture.	UC16 (Protect data)
SEC-AUTH-001	Essential	IT-L0-C3, IT-L1-4	QUAL-SEC-001	The system shall enforce mutual authentication for system-to-system communications using X.509 certificates .	Test: Reject invalid/expired certificates; accept valid certificates only.	UC16 (Mutual authentication)
SEC-AUD-001	Essential	IT-L0-C3, BO-L0-C4	QUAL-SEC-001, QUAL-TRC-001	The system shall maintain audit logs for access to operational data and for administrative configuration changes, including actor identity and timestamps.	Inspection/Test: Perform actions; confirm corresponding audit records.	UC17 (Audit access)

5 Technology Requirements (Latest Tech)

The following technology requirements define the intended operating platform, data store, and interface patterns selected by Brigman Deep Sea Technology to satisfy the Project Specification constraints.

Tech ID	Area	Specification
TECH-PLAT-001	Platform	Vehicle-side software components shall run on a Linux-based OS with ROS2 middleware support; mothership-side services shall support deployment on standard workstation/server environments.
TECH-DB-001	Database	The system shall store operational and mission records in a structured database that supports retention policies and encrypted storage (e.g., PostgreSQL or equivalent).
TECH-NET-001	Networking	The system shall support message transport compatible with constrained links (DDS where available) and shall support store-and-forward buffering during outages.
TECH-API-001	APIs	External and internal service interfaces shall use versioned JSON schemas and provide a documented API contract for integration and export.

Table 8: Technology requirements for operating environment and platform

6 Quality Development Model

Brigman Deep Sea Technology applies the **BDST Quality Model v1.0** to ensure requirements are not only functional but also safe, secure, and verifiable under operational constraints. Each requirement in Section 4 is tagged with one or more quality feature IDs below.

Quality ID	Feature Name	What it Covers / Measures
QUAL-SAF-001	Safety Integrity	Measures correct detection of abnormal conditions, safe-state transitions, and recovery behavior under fault injection.
QUAL-SEC-001	Security & Governance	Measures encryption coverage, authentication strength, audit completeness, and unauthorized access resistance.
QUAL-RES-001	Resilience under Constraints	Measures mission continuity and safe behavior during comm outages, bandwidth limits, and sensor degradation.
QUAL-TRC-001	Traceability & Accountability	Measures ability to reconstruct mission timelines and decisions from logs with actor attribution and retention.
QUAL-OPS-001	Operability & Performance	Measures operator ability to plan missions, interpret telemetry/alerts, intervene reliably, and meet timing/performance thresholds (telemetry update rates, detection latency, report generation, retrieval times).

Table 9: BDST Quality Model v1.0 (quality feature names and measurement intent)

7 Integrated Requirements Traceability Matrix (iRTM)

The iRTM traces each Project Specification clause to the corresponding SRS requirement(s) or SRS section(s), using minimized paraphrases to preserve CReAM meaning while remaining concise.

Table 10: iRTM: Project Specification (PS) to SRS traceability

iRTM ID	PS Ref	PS Clause (min.)	SRS Element	SRS Clause / Coverage (min.)
RTM-001	BO-L0-C1	Unmanned operations at extreme depths.	NAV-EXEC-001	Autonomous execution without continuous human control.
RTM-002	BO-L0-C2	Safety-first: eliminate human exposure.	SAF-STATE-001; SYS-TERM-001	Safe-state bias and orderly termination under risk.
RTM-003	BO-L0-C3	Extreme pressure/low visibility/limited comm.	COM-DEG-001; SAF-EMG-001	Constrained-link behavior; abnormal detection.
RTM-004	BO-L0-C4	Comply with maritime safety/data governance.	SEC-AUD-001; DATA-LOG-002	Auditable access and retained records.
RTM-005	BO-L0-A1	Missions supervised by trained personnel.	Section 2.7	Operational assumption recorded (software supports supervision).
RTM-006	BO-L1-1	Autonomous mission execution required.	NAV-PLAN-001; NAV-EXEC-001; NAV-WPT-001	Plan, autonomous execution, and waypoint performance.

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iRTM ID	PS Ref	PS Clause (min.)	SRS Element	SRS Clause / Coverage (min.)
RTM-007	BO-L1-2	Remote mission oversight required.	COM-MON-001; COM-INT-001	Telemetry updates and remote interventions.
RTM-008	BO-L1-3	Coordinated multi-submersible operation.	NAV-FLEET-001	Concurrent supervision and mission assignment.
RTM-009	BO-L1-4	Handle abnormal/emergency conditions.	SAF-EMG-001..003; SAF-STATE-001	Detection thresholds and safe-state transition.
RTM-010	BO-L1-5	Continue safely or terminate orderly.	SYS-CONT-001; SYS-TERM-001; SYS-REC-001	Degraded continuation, termination, recovery coordination.
RTM-011	BO-L1-6	Accountability for mission activities/decisions.	DATA-LOG-001; SEC-AUD-001	Operator-attributed logging and auditability.
RTM-012	BO-L1-7	Outputs usable for ops and analytics.	DATA-LOG-003; DATA-LOG-004	Exportable records and summary reporting.
RTM-013	BO-L2-4.1	Abnormal conditions identifiable.	SAF-EMG-001..003	Defined abnormal thresholds and detection timing.
RTM-014	BO-L2-4.2	Transition to safe state on critical.	SAF-STATE-001	Defined safe operational state behavior.
RTM-015	BO-L2-6.1	Activities/decisions traceable.	DATA-LOG-001; SEC-AUD-001	Trace logs and audit logs with actor identities.
RTM-016	BO-L2-6.2	Records support review/investigation.	DATA-LOG-002..004	Retention, export, and post-mission reporting.
RTM-017	IT-L0-C1	Remote supervision/control context.	COM-MON-001; COM-INT-001	Remote monitoring and intervention functions.
RTM-018	IT-L0-C2	Constrained bandwidth/latency/intermittent.	COM-DEG-001	Explicit bandwidth, latency, and outage thresholds.
RTM-019	IT-L0-C3	Protect operational data per obligations.	SEC-ENC-001; SEC-AUTH-001; SEC-AUD-001	Encryption, authentication, audit.
RTM-020	IT-L0-A1	Customer provides sensors for integration.	INT-SENS-001	ROS2/DDS/JSON-based sensor integration.
RTM-021	IT-L1-1	Remote monitoring capability required.	COM-MON-001	Telemetry update rate requirement.
RTM-022	IT-L1-2	Remote intervention capability required.	COM-INT-001	Remote command/ack requirement.
RTM-023	IT-L1-3	Multi-vehicle supervision required.	NAV-FLEET-001	3-vehicle concurrent supervision.
RTM-024	IT-L1-4	Protect confidentiality/integrity of data.	SEC-ENC-001; SEC-AUTH-001	Encryption and mutual authentication.
RTM-025	IT-L1-5	Logging mission events and operator actions.	DATA-LOG-001..004	Logging, retention, export, reporting.

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iRTM ID	PS Ref	PS Clause (min.)	SRS Element	SRS Clause / Coverage (min.)
RTM-026	IT-L1-6	Safe operation under degraded comm.	COM-DEG-001; SYS-CONT-001	Degraded comm handling and mission continuity.
RTM-027	IT-L2-5.1	Logs sufficient for review/accountability.	DATA-LOG-003; DATA-LOG-004	Exportable records and review-ready reports.

8 Change Mitigation Protocol

This protocol reduces downstream SDLC errors by preventing uncontrolled requirement drift, ensuring every change is evaluated for impact on implementation, verification, and traceability.

Change ID	Control / Strategy
CHG-CTRL-001	Baseline control: the SRS is baselined at Version 1.0; all changes require a change request (CR) recorded in the repository.
CHG-CTRL-002	Change classification: changes are classified as Editorial, Minor (no external impact), or Major (scope/safety/security/performance impact).
CHG-CTRL-003	Impact analysis: each CR must list affected requirement IDs, affected iRTM entries, verification updates, and schedule/risk impacts before approval.
CHG-CTRL-004	Approval gates: Major changes require approval by the Customer-designated approver and the Vendor authenticator; Minor changes require Vendor authenticator approval.
CHG-CTRL-005	Traceability maintenance: any approved change must update Section 4 and Section 7 in the same revision to keep the iRTM consistent.
CHG-CTRL-006	Verification alignment: changes that affect timing, safety, or security requirements must include updated test cases and acceptance criteria prior to implementation.

Table 11: Change mitigation controls (codified)

9 Addenda

Addendum ID	Statement
ADD-001	Project direction baseline: This SRS assumes the customer direction change to a fully unmanned, autonomous-first operating model is in force for all missions.
ADD-002	Cost redaction note: Any cost figures referenced from the Project Specification are treated as indicative and may be redacted in released versions to avoid biasing proposals.

Table 12: Addenda (purpose-appropriate notes adopted from the Project Specification)

10 Requirements Modeling

The following models are included to support stakeholder understanding and to maintain traceability between requirements and operational context. Refer to Figure 2 (Use Case Diagram) for the consolidated actor–system interactions used throughout this SRS.

Additional Project Specification models (e.g., organization chart and roles/responsibilities) are adopted by reference from the Project Specification appendices to avoid divergence across artifacts.

11 Sign-off

11.1 Vendor Authentication

Name	Employee ID	Signature
Muhammad Ridwan Putra Jasni (Authenticator)	BDST-RE-BA-001	

Table 13: Vendor authenticator sign-off (signature provided)

11.2 Designated Approvers (No Signatures Required)

Organization	Role	Name (Designated)
Customer (Triton Deepwater)	Project Leader / Requirements Approver	Sarah Whitman
Customer (Triton Deepwater)	Safety & Compliance Approver	(TBD)
Vendor (Brigman)	Engineering Delivery Lead	(TBD)
Vendor (Brigman)	Verification & Validation Lead	(TBD)

Table 14: Designated approvers (placeholders where not yet nominated)

Repository Information

PROJECT REPOSITORY
https://github.com/Spooky312/The-Abyss-Navigation-Suite-INF2113 <i>SRS, models, and traceability artifacts maintained under version control</i>

A Glossary

- **BCIC:** Breakdown, Clarify, Interpret, Categorize.
- **CRaM:** Correct, Relevant, and Measurable requirements organization principle.
- **iRTM:** Integrated Requirements Traceability Matrix.
- **Safe-state:** neutral buoyancy, minimal power, beacon active.

B To Be Determined (TBD) List

- Customer nomination of Safety & Compliance approver name.
- Final selection of database product/version for TECH-DB-001 (if different from PostgreSQL-equivalent).
- Final JSON schemas for INT-SENS-001 (to be published with the Interface/API guide).

C Figures

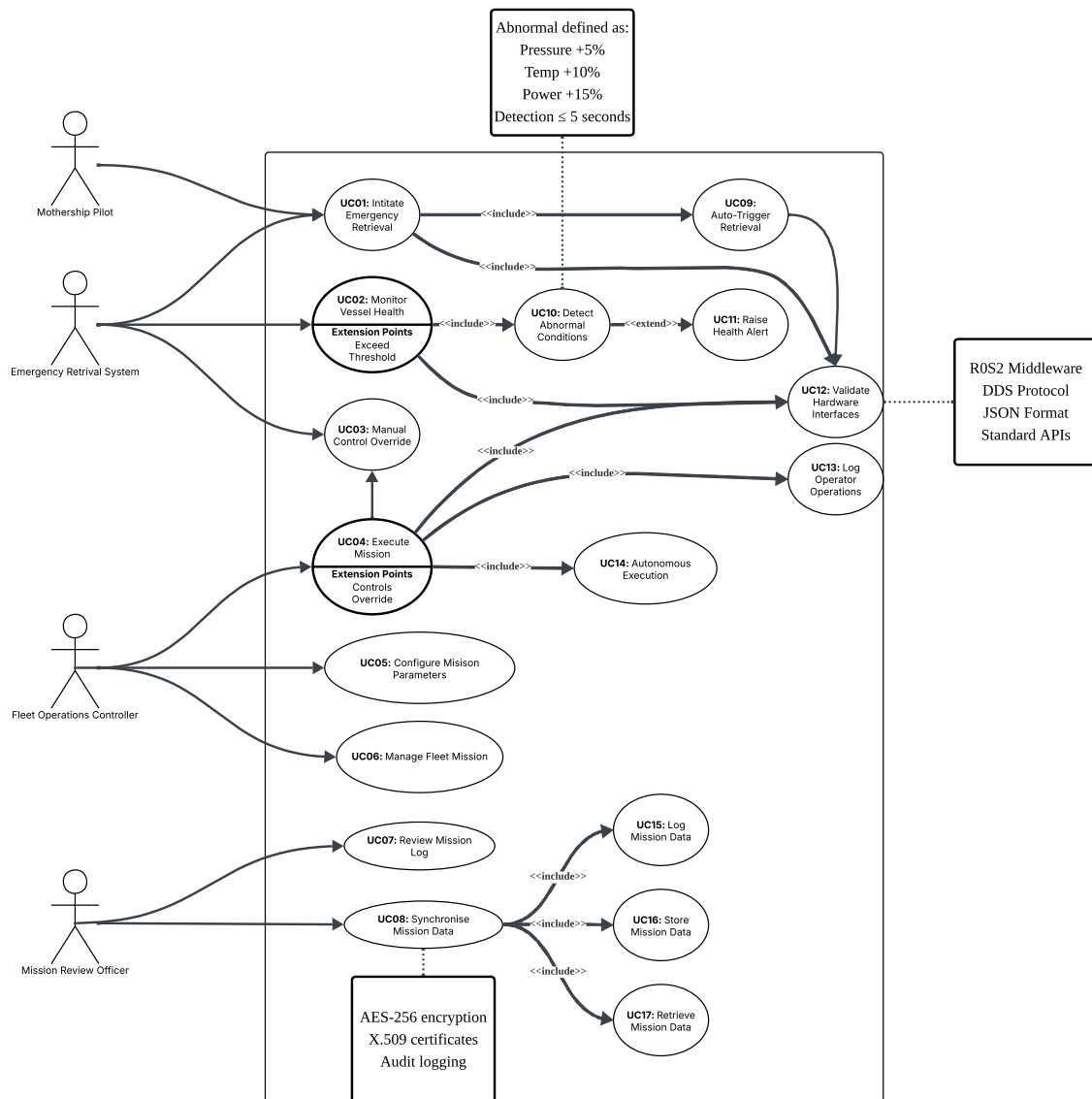


Figure 2: Use Case Diagram

Instructions

The diagram illustrates the primary actors (surface controller, mothership crew, autonomous submersible) and their interactions with major system functions such as mission planning, launch/recovery, sensor telemetry, fault handling, and data logging. Follow each actor's swim lane to see how the mission progresses from preparation to post-mission data review; the colored connectors highlight safety-critical flows (emergency surfacing, manual override) that must remain available throughout operations.

Appendix A: Response Sequence

Use Case No.	Use Case Flow (Description)	Use Case End Goal
UC01 – Initiate Emergency Retrieval	Main Flow: 1. Pilot detects an emergency situation. 2. Pilot selects the emergency retrieval command. 3. System validates hardware interfaces. 4. System activates retrieval protocol. 5. Vessel begins return-to-surface procedure.	The vessel is safely retrieved during an emergency.
UC02 – Monitor Vessel Health	Main Flow: 1. System continuously receives vessel telemetry. 2. System checks pressure, temperature, and power levels. 3. System compares readings with safety thresholds. 4. System detects abnormal readings if thresholds are exceeded. 5. System prepares alerts for operators.	Vessel health is monitored continuously to detect issues early.
UC03 – Manual Control Override	Main Flow: 1. Operator selects the manual override option. 2. System verifies operator authorization. 3. System disables autonomous control temporarily. 4. Operator sends manual navigation commands.	Operator gains direct control of the vessel when necessary.
UC04 – Execute Mission Controls Override	Main Flow: 1. System receives manual control commands. 2. System validates command parameters. 3. System applies override to mission controls. 4. System logs operator actions. 5. Vessel executes the override instructions.	Mission continues under manual control safely.
UC05 – Configure Mission Parameters	Main Flow: 1. Fleet Controller opens mission configuration interface. 2. Controller enters mission parameters (depth, path, duration). 3. System validates parameter limits. 4. Controller confirms configuration. 5. System stores mission parameters.	Mission parameters are prepared for safe mission execution.
UC06 – Manage Fleet Mission	Main Flow: 1. Fleet Controller views fleet dashboard. 2. Controller assigns missions to vessels. 3. System distributes mission data to vessels. 4. Controller monitors fleet progress. 5. System updates fleet status in real time.	Multiple vessels are coordinated efficiently.
UC07 – Review Mission Log	Main Flow: 1. Mission Review Officer accesses mission log system. 2. Officer selects a completed mission. 3. System retrieves stored mission records. 4. System displays full mission timeline.	Mission performance is evaluated using stored logs.
UC08 – Synchronise Mission Data	Main Flow: 1. System collects mission data from vessels. 2. System encrypts data using secure protocols. 3. System synchronizes data with control station servers. 4. System verifies successful synchronization. 5. System updates mission data repository.	Mission data remains consistent across all systems.

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Use Case No.	Use Case Flow (Description)	Use Case End Goal
UC09 – Auto-Trigger Retrieval	Main Flow: 1. System detects abnormal operating conditions. 2. System evaluates emergency thresholds. 3. System automatically triggers retrieval protocol. 4. Vessel receives retrieval command. 5. Vessel begins safe recovery procedure.	Automatic safety retrieval occurs during critical failures.
UC10 – Detect Abnormal Conditions	Main Flow: 1. System continuously monitors sensor data. 2. System evaluates pressure, temperature, and power metrics. 3. System compares metrics to defined limits. 4. System identifies abnormal condition. 5. System notifies alert subsystem.	Abnormal system conditions are detected quickly.
UC11 – Raise Health Alert	Main Flow: 1. System receives abnormal condition signal. 2. System generates health alert notification. 3. Alert is displayed on operator dashboard. 4. System logs the alert event. 5. Operator reviews and decides next action.	Operators are notified of system health issues.
UC12 – Validate Hardware Interfaces	Main Flow: 1. System checks connected hardware modules. 2. System tests communication interfaces. 3. System verifies sensor availability. 4. System confirms interface compatibility. 5. System reports validation status.	Hardware readiness is verified before operations.
UC13 – Log Operator Operations	Main Flow: 1. Operator performs system action. 2. System captures action details. 3. System records timestamp and user identity. 4. System stores log entry securely. 5. Audit log is updated.	Operator activities are recorded for auditing.
UC14 – Autonomous Execution	Main Flow: 1. System loads configured mission parameters. 2. Vessel begins autonomous navigation. 3. System monitors execution progress. 4. System records telemetry data. 5. Mission continues until completion or interruption.	Vessel completes mission autonomously.
UC15 – Log Mission Data	Main Flow: 1. System collects mission telemetry data. 2. System records sensor readings. 3. System timestamps each data entry. 4. Data is formatted for storage. 5. Log records are prepared for database storage.	Mission telemetry data is recorded during operations.
UC16 – Store Mission Data	Main Flow: 1. System receives mission log records. 2. System encrypts stored data. 3. System writes records to database storage. 4. System confirms successful storage. 5. Data repository is updated.	Mission data is securely stored in the system database.

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Use Case No.	Use Case Flow (Description)	Use Case End Goal
UC17 – Retrieve Mission Data	Main Flow: 1. Officer requests mission data retrieval. 2. System searches mission database. 3. System retrieves requested data. 4. System formats data for display. 5. Data is presented to the officer.	Stored mission data becomes accessible for analysis.

Appendix B: Wireframe